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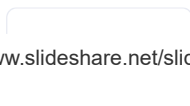
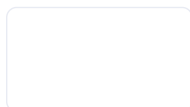
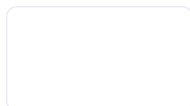
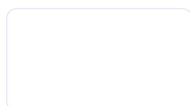
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1. CHROMIUM GROUP

2. **GROUP 6-CHROMIUM GROUP ELEMENTS** – CHROMIUM ($^2\text{CR}[\text{AR}]3\text{D } 4\text{S}^1 4^5$ MOLYBDENUM (^2MO)- [$^4\text{KR}]4\text{D } 5\text{S}^1 5$ TUNGSTEN (W)- [$\text{XE}]4\text{F}^1 5\text{D } 6\text{S}^2 7444$ 3. • **Chromium is** the twenty – first most abundant element by weight in the earth's crust. • Molybdenum and tungsten are quite rare. ABUNDANCE ELEMENT PPM RELATIVE ABUNDANCE Cr 122 21 Mo 12 56 W 12 56 Abundance of the elements in the earth's crust by weight4. **DISCOVERY OF CHROMIUM Chromium** was discovered by louis nicolas vauquelin (french chemist) in 1798. He was intrigued by a bright red mineral that had been discovered in a Siberian goldmine in 1766. it was referred to as Siberian red lead and known as crocoite(PbCrO_4)5. • **The element** chromium is named after the Greek word 'chroma' which means colour. • The only commercially important ore of Cr is chromite $\text{FeCrO}_2 \cdot 4$ • Chromite has a slight lustre and looks like pitch, with a brownish cast to the colour It may be slightly magnetic6. **EXTRACTION OF CHROMIUM Chromite $\text{FeCrO}_2 \cdot 4$** NaOH(l) air oxidation $\text{NaCrO}_2 \cdot 4$ O leach Aq. $\text{NaCrO}_2 \cdot 4$ evaporate Solid $\text{NaCrO}_2 \cdot 4$ C, heat reduction $\text{CrO}_2 \cdot 3$ Si or Al ,heat Cr7. **Extraction of chromium**8. **USES Chromium is produced** in two forms ferrochrome and pure Cr metal, For Ferrochrome is an alloy containing Fe, Cr and C $\text{FeCrO} + \text{C Fe} + 2\text{Cr} + 4\text{CO}_2 \cdot 4$ It is used to make many Ferrous alloys, including stainless steel and hard Chromium steel. Ferrochrome10. **MOLYBDENUM It was discovered** by Carl Wilhelm Scheele in 1778 It was 1st isolated by Swedish chemist Peter Jacob Hjelm in 1781 The name molybdenum comes from Greek word molybdos which mean "lead"11. **OCCURANCE AND EXTRACTION Molybdenum** occurs as the mineral molybdenite MoS_2 MoS is roasted in air, converting it to MoO_3 MoO may be heated with Fe and Al to give ferromolybdenum , which is then added to steel. 1. Pure Mo is obtained by dissolving MoO in dilute $\text{NH}_3 \cdot 4\text{OH}$ and precipitating ammonium molybdate, dimolybdate or paramolybdate. This is reduced with hydrogen to give the metal Mo12. **USES Almost 90% of** Mo is used to make cutting steel or stainless steel. Mo metal is used as a catalyst in the petrochemical industry.14. **TUNGSTEN It was discovered** by "juan and faustro elhuyar" in spain at 1783 The symbol of the element is "W" The w stands for wolfram Tungsten is a rare, naturally occurring silvery-gray metal15. **OCCURANCE AND EXTRACTION Tungsten** occurs as tungstates, the most common being wolframite FeWO_4 . MnWO_4 and scheelite CaWO_4 Different processes are used to extract W from wolframite and scheelite. Wolframite is fused with $\text{NaCO}_3 \cdot 3$, forming sodium tungstate, which is leached out and acidified to give 'tungstic acid' (the hydrated oxide). Scheelite is acidified with HCl when 'tungstic acid' is precipitated and other materials dissolve. 'Tungstic acid' is then heated to give the anhydrous oxide, which is reduced to give the metal by heating with hydrogen at 850°C .16. **USES • Both Mo and W** are alloyed with steel, giving very hard alloys Which are used to make cutting steel. • This is used to make machine tools. • Cutting steel retains its cutting edge even when the metal becomes red hot18. **Thank you**

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